



~ Internship Technical Report ~

DAY 02

Wi-Fi Security Checker

Internship Record | VIEH GROUP



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Internship Program	VIEH – 30-Day Cybersecurity Automation Series
Task Title	Wi-Fi Security Checker
Tools & Technologies	Python, subprocess, Network Commands
Operating Platform	Kali Linux & Windows



🌸 Project Goal

The objective of this project is to design a Wi-Fi Security Checker using Python that examines the current network connection, identifies wireless security details, and performs a basic security assessment of the connected network environment. The utility was developed to strengthen understanding of wireless networking concepts, operating system networking commands, and Python automation techniques.

Since the project was initially implemented inside a Kali Linux VirtualBox environment, direct wireless hardware access (`wlan0`) was unavailable. Therefore, the tool was adapted to function using the virtual Ethernet interface (`eth0`) provided by the virtual machine. To further validate wireless functionality, the project was additionally executed on Windows, where the system's native Wi-Fi adapter and wireless networking support were available.

🌸 Overview

The Wi-Fi Security Checker is a Python-based cybersecurity utility developed to inspect the active network connection and evaluate its security posture. The project focuses on gathering network-related information such as the connected network name, signal strength, encryption type, and general security analysis using operating system networking commands executed through Python.

Initially, the project was tested within Kali Linux running inside VirtualBox. Due to virtualization limitations, the virtual machine primarily exposed a virtual Ethernet adapter (`eth0`) instead of a physical wireless interface (`wlan0`). As a result, the Linux implementation was modified to operate correctly within the virtual networking environment while still demonstrating concepts such as network enumeration, interface detection, subprocess automation, and security analysis.

🌸 Core Capabilities

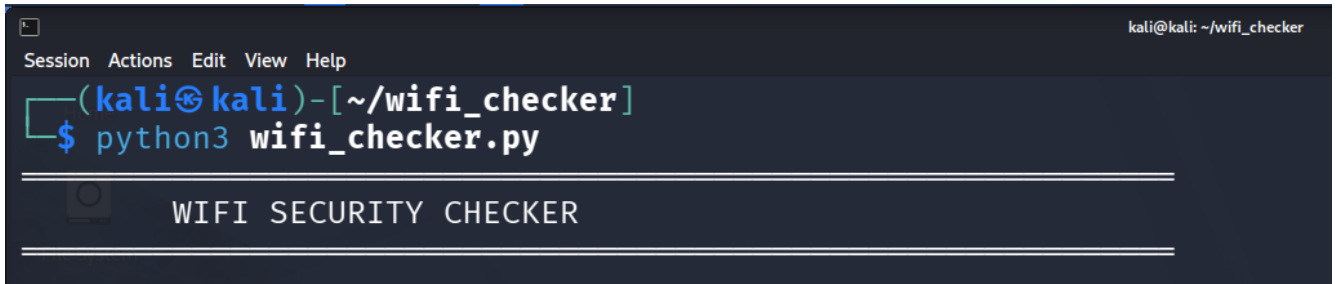
- 🌸 Detection of Active Network Interfaces
- 🌸 Wi-Fi Security Enumeration
- 🌸 Encryption Type Identification
- 🌸 Signal Strength Analysis
- 🌸 Cross-Platform Compatibility (Linux & Windows)
- 🌸 Virtual Machine Network Detection
- 🌸 Beginner-Friendly Python Automation
- 🌸 Lightweight & Easy to Execute

🌸 Tech Stack

- 🌸 Python 3
- 🌸 subprocess Module
- 🌸 platform Module
- 🌸 socket Module
- 🌸 Windows Networking Commands (`netsh`)
- 🌸 Linux Networking Commands (`nmcli`)
- 🌸 Kali Linux, Windows OS, VirtualBox Virtualization Environment

🌸 Visual Demonstrations

Screenshot 1 – Running the Wi-Fi Security Checker on Kali Linux



A terminal window titled 'kali@kali: ~/wifi_checker' with a menu bar (Session, Actions, Edit, View, Help). The prompt is '(kali@kali)-[~/wifi_checker]' and the command '\$ python3 wifi_checker.py' has been entered. The output shows 'WIFI SECURITY CHECKER' with a small icon to the left, separated by horizontal lines.

Screenshot 2 – Network Interface Detection using eth0 in VirtualBox



The output of the script is displayed in a terminal window. It shows system information, network device details, security status, and system info.

```
System Name      : kali
IP Address       : 127.0.1.1

[ NETWORK DEVICES ]
DEVICE  TYPE      STATE      CONNECTION
eth0    ethernet  connected  Wired connection 1
lo      loopback  connected (externally)  lo

[ SECURITY STATUS ]
[+] Ethernet connection detected
[+] Stable and secured virtual connection
[!] No real wireless adapter detected
[!] Running in Virtual Machine environment

[ SYSTEM INFO ]
Operating System : Linux

[v] Scan Completed Successfully
```

Screenshot 3 – Running the Wi-Fi Security Checker on Windows

```
C:\Windows\System32>python --version
Python 3.14.3

C:\Windows\System32>mkdir wifi_checker

C:\Windows\System32>cd wifi_checker

C:\Windows\System32\wifi_checker>notepad wifi_checker.py

C:\Windows\System32\wifi_checker>python wifi_checker.py
=====
                WIFI SECURITY CHECKER
=====
```

Screenshot 4 – Wireless Network Information Detected on Windows

```
=====
                WIFI SECURITY CHECKER
=====

System Name : MubarizZ
IP Address  : 192.168.0.106
Operating System : Windows

[ WINDOWS WIFI INFO ]
Connected Wi-Fi : Mubariz
Signal Strength : 93%
Authentication  : WPA2-Personal

[ SECURITY STATUS ]
[+] Network enumeration successful

=====
[✓] Scan Completed Successfully

C:\Windows\System32\wifi_checker>
```

🌸 How It Operates

- ✿ The program first identifies the operating system platform.
- ✿ On Linux systems, the tool executes `nmcli` commands to enumerate active network devices and connection status.
- ✿ In the VirtualBox environment, the tool detects the virtual Ethernet interface (`eth0`) instead of a physical Wi-Fi adapter.
- ✿ On Windows systems, the tool uses `netsh wlan show interfaces` to retrieve wireless network details.
- ✿ The program extracts information such as SSID, signal strength, and encryption/authentication type.
- ✿ A basic security assessment is then displayed based on the detected encryption standard.
- ✿ Finally, all collected information is presented in a structured and user-friendly format.

✿ Sample Output

Environment	Output Summary
Kali Linux	Ethernet connection detected through <code>eth0</code>
Windows	Connected Wi-Fi detected with SSID, signal, WPA2/WPA3 details

✿ Key Takeaways

- ✿ Fundamentals of Wireless Networking
- ✿ Python subprocess Automation
- ✿ Cross-Platform Python Scripting
- ✿ Linux Network Enumeration
- ✿ Windows Wireless Networking Commands
- ✿ Virtual Machine Networking Concepts
- ✿ Security Assessment Logic

This project is intended strictly for educational learning and authorized security analysis purposes only.

Unauthorized network monitoring or wireless assessment activities performed without proper permission may violate legal and ethical regulations.